

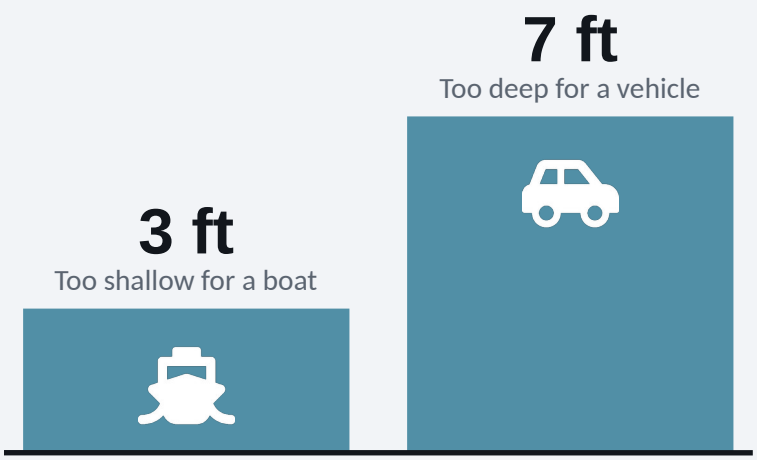
PROJECT ARGO · TEAM TRICHOKRO

The amphibious ambulance and rescue vehicle built in Bangladesh, for a drowning delta

“In water, in land — your life is in our hand.”

ARKGO

When the water rises, the roads disappear



Flood depth changes street to street — no single boat or vehicle fits.

Haors, chars & swamps

Half land, half water for months every year.

Southern Bangladesh

Patients are carried to boats, then re-loaded onto vehicles.

Waterlogged cities

Dhaka and Chattogram flood after routine downpours.

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2024 floods, zero water ambulances

5.8M
people hit by the Aug-2024 flash floods

1M+
people completely cut off by water

502K
displaced into 3,403 shelters

75%
of Sylhet district under water

~800
government ambulances for 170M people

0
water ambulances in the public fleet



Sources: UN OCHA Sitrep No. 02 (Aug 2024); ReliefWeb; NDRCC; DGHS

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Meet ARKGO — one vehicle, both worlds



RESCUE AMBULANCE RELIEF

Ambulance & rescue vehicle

Stretcher, oxygen, up to 11 seats

Electric on land

5 kW EV drive, 40 km/h

ICE engine in water

Independent, with guarded prop

Solar-assisted

~500 W roof panel, ~2 days autonomy

ARKGO

Two drivelines, one sealed hull

1

LAND — ELECTRIC

- 5 kW motor, rear axle
- 10 kWh LiFePO4 in the floor
- 40 km/h · 10 in clearance

2

TRANSITION

- Drive straight in
- Sealed hull takes the load
- Tires add ~155 kg of lift

3

WATER — ICE ENGINE

- ICE engine + guarded prop
- 10 km/h in debris-laden water
- Wheels act as roll sponsons

SEAMLESS No ramps, no cranes, no patient transfer — the same cabin works on land and water.

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Built for the flood, from every angle



Water mode: the guarded stern propeller is steered from the rear platform.

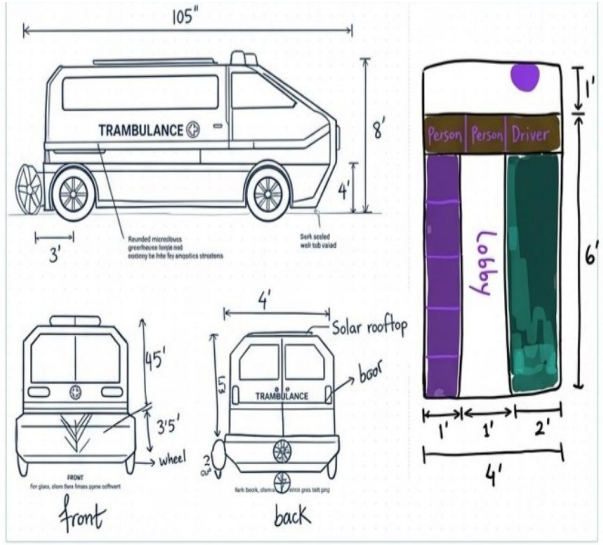




Rear access: roll-in stretcher loading.

Technical specification sheet

Overall length	105 in (8.75 ft)
Overall height	8 ft (hull tub 3 ft)
Cabin footprint	4 ft × 6 ft
Ground clearance	10 in
Wheels	24 in OD, buoyant
Land drive	5 kW electric · 40 km/h (≈ 4 kW needed)
Land range	≈ 100 km per charge
Water drive	ICE engine, 10–15 hp · guarded prop · 10 km/h
Battery	LiFePO4 · 10 kWh (48 V, 200 Ah)
Solar	Rooftop panel, ~500 W
Capacity	3 front + 4 bench + stretcher or 4 seats
Max load	≈ 2,000 kg
Chassis	Welded steel truss space-frame



Rev-2 engineering views & dimensions

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Three energy sources, ~2 days autonomy

BATTERY



10 kWh

LiFePO4 · 48 V · 200 Ah, long-life chemistry

SOLAR



of a full charge refilled per sunny day

~500 W

roof panel ≈ 2 kWh/day, keeps radios and devices alive

ICE ENGINE

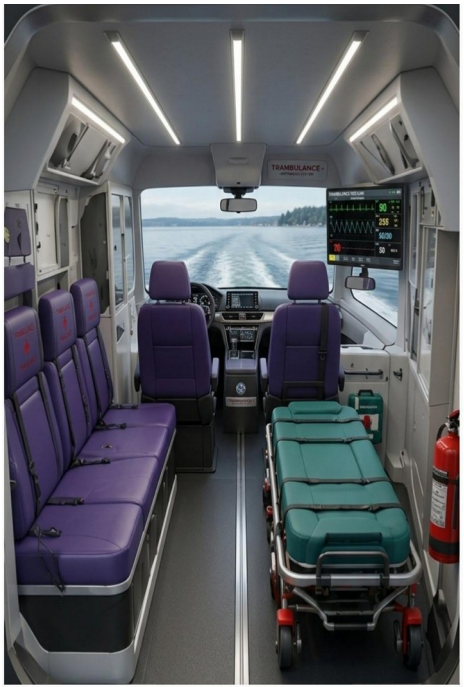


10–15 hp

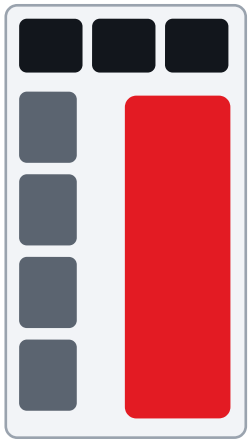
independent water drive — never depends on the battery

≈ 2 days of autonomy · ≈ 100 km land range · +~20 km/day from solar

Inside: an ER that floats

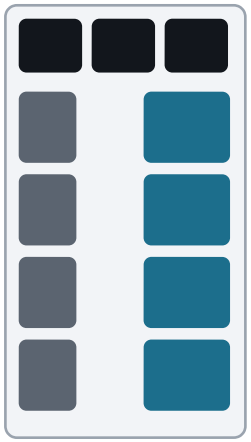


AMBULANCE



7 seated + 1 patient

RESCUE



11 seated

Primary medical equipment

Monitor, IV points, first-aid

Oxygen cylinder

Fixed mount beside the stretcher

8 life vests + throw ring

One for every rear seat

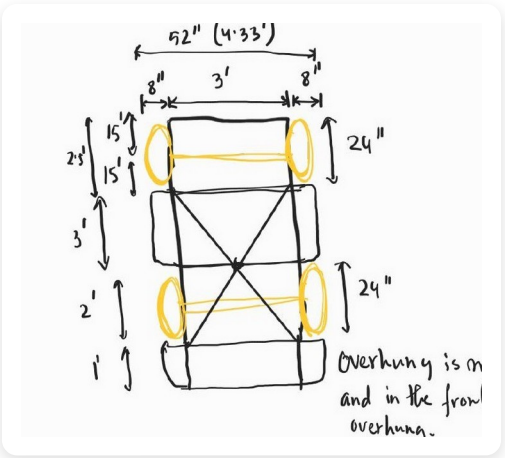
Fire extinguisher

Crash-rated stowage

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Light truss chassis, 2-tonne max load

MAX LOAD CAPACITY · ≈ 2,000 kg



≈ 17 in — draft at a 1.2 t mission load

≈ 19 in — freeboard at 1.2 t — about 2× the minimum

+155 kg — reserve lift from four air-filled tires

10 in — freeboard still kept at the full 2 t

PASS — IT FLOATS

Welded steel truss (~120 kg), foam-filled hull voids.

ARKGO

Feasibility scorecard: every check passes

Buoyancy at 1.2 t — PASS
≈ 17 in draft, ≈ 19 in freeboard

Max load 2 t — PASS
10-in freeboard, incl. tire lift

Land speed 40 km/h — PASS
5 kW motor; ≈ 4 kW needed

Water speed 10 km/h — PASS
10–15 hp ICE, guarded prop

Energy autonomy — PASS
10 kWh + 500 W solar, ~2 days

Corrosion life — PASS
FRP hull, galvanized truss

Verdict: fully feasible — 40 km/h on land, 10 km/h in water, up to 2 t of load. Zero showstoppers.

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Safety systems & certification path

Unsinkable by design
Foam-filled voids + buoyant tires

Guarded propeller
Cage keeps limbs, ropes and hyacinth out

Twin bilge pumps
Auto + manual; lip-sealed stub axles

IP67 electrics
Sealed battery box, master kill switch

Rescue kit
8 life vests, throw ring, VHF, strobes

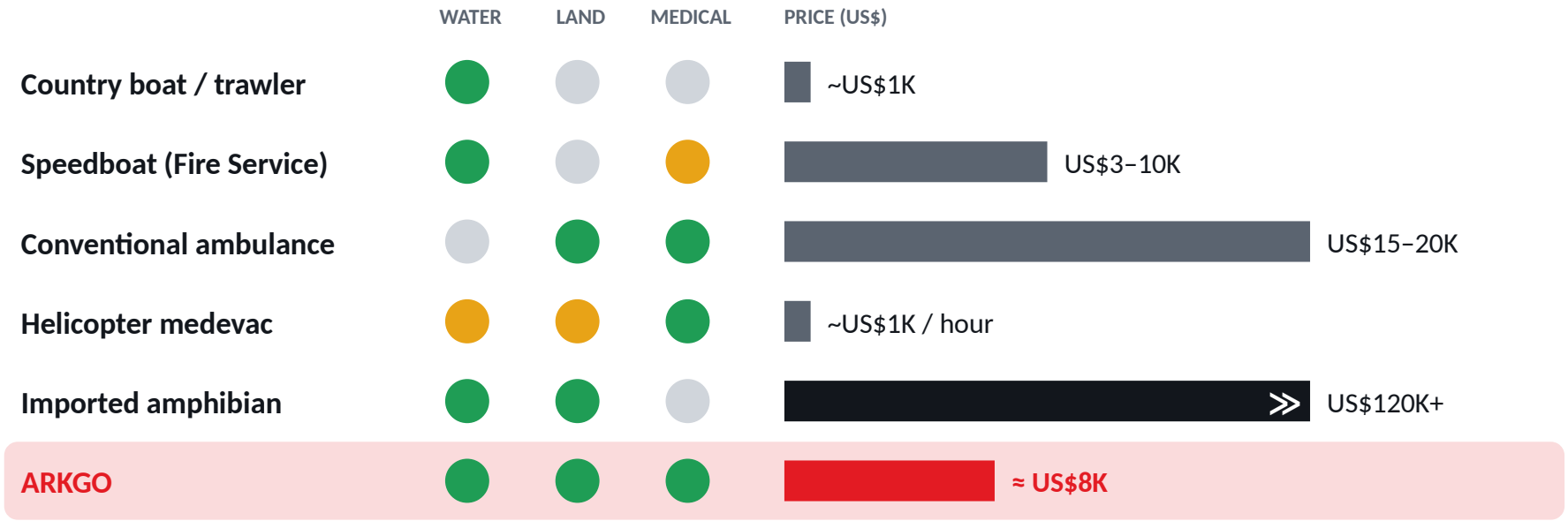
Fire safety
ABC extinguisher, fuel shutoff, fusing

CERTIFICATION PATH

BRTA special-purpose registration → Dept. of Shipping small-craft survey → BSTI component standards → CE small-craft (export markets)

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No rival covers the whole rescue chain



15× cheaper than imported amphibians — and below a road ambulance

Competitor pricing: public list prices / operator rates.

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Five sustainable points

01

Radical price point

≈ US$8K — below a road ambulance, 1/15 of an imported amphibian

02

Made in Bangladesh

~70% local content, local welders and parts

03

Medical-first design

Built around the stretcher, not retrofitted

04

Dual-energy resilience

Solar + battery + ICE: ~2 days of autonomy, long-life LiFePO4

05

Climate-finance ready

EV drive + adaptation mission fit GCF and World Bank funding

Priced for upazila budgets, not defense budgets.

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Business model canvas

KEY PARTNERS

Fire Service & Civil Defence · DGHS · light-engineering workshops · battery & motor importers · BDRCS & NGOs

KEY ACTIVITIES

Truss fabrication · FRP hull molding · powertrain & medical fit-out

KEY RESOURCES

Rev-2 design IP · weld & mold shop · certification dossier

VALUE PROPOSITION

The only ambulance that keeps driving when roads flood — at US$8K, serviceable in any bazar

RELATIONSHIPS

Long-term AMC · operator training · 24/7 parts hotline

CHANNELS

Govt e-tenders (e-GP) · NGO procurement · G2G · export distributors

CUSTOMER SEGMENTS

B2G: DGHS, Fire Service, city corporations, forces

B2B: NGOs, private hospitals

Communities via donors

COST STRUCTURE

COGS ≈ 77% at launch → 68% at scale · fixed: shop lease, engineering, certification

REVENUE STREAMS

Unit sales · 8%/yr AMC · EMS-as-a-service leasing · export CKD kits & licensing

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TAM: ~11,000 units worldwide · ~US$88M

TAM ~11,000 units · US$88M

Every flood-exposed EMS buyer in the whole world

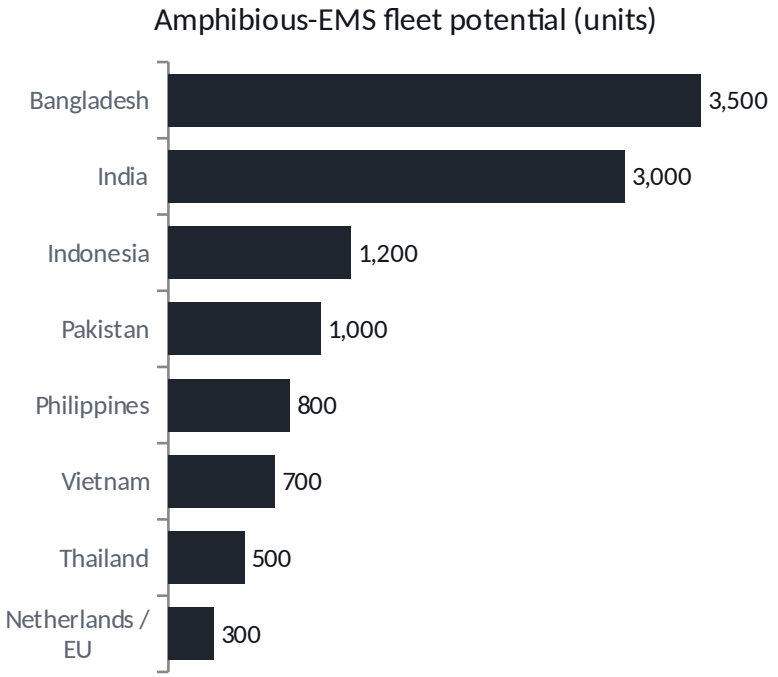
SAM 3,500 units · US$28M

Every flood-exposed EMS buyer in Bangladesh

SOM 120 units · ~US$1M

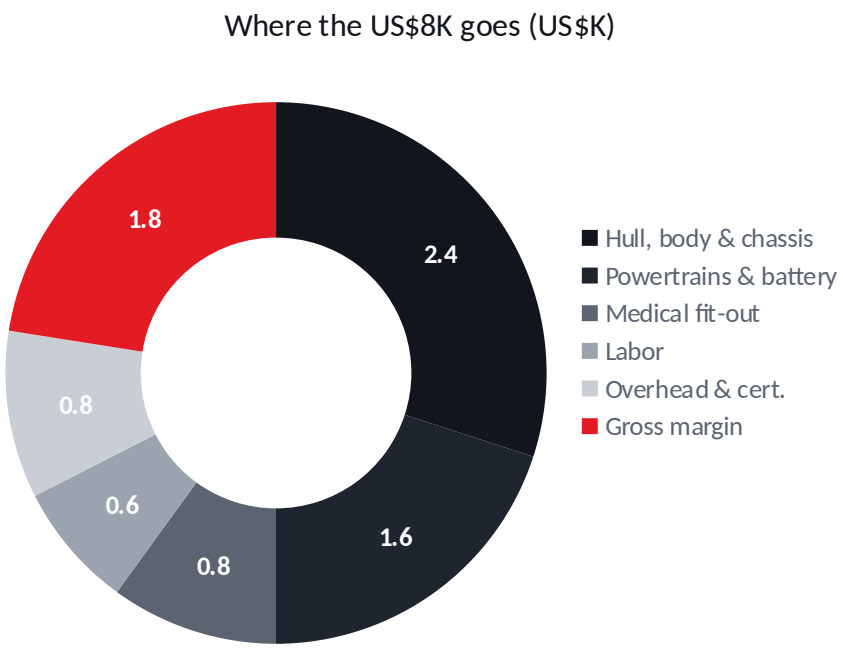
3 years: pilots, Fire Service, 2 NGO fleets

At US$8K per unit · plus 8%/yr AMC on the installed fleet



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Profitable from unit one at US$8K



US$8K

unit price

22.5%

gross margin at launch, ~32% at scale

US$640

yearly AMC per unit (8%)

US$240

monthly lease, payback < 3 years

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Prove in one monsoon, scale on tenders

1

MONTH 1–9

Prove

2 pilot units with Fire Service and a Sylhet upazila

2

MONTH 9–18

Win B2G

Pilot data into DGHS and Fire Service e-GP tenders

3

YEAR 2

Service

Leased units and trained crews, linked to 999

4

YEAR 3

Export

Climate-finance deals across flood-prone Asia

PRIORITY DEPLOYMENT ZONES · PHASE 1 IN RED

- I Cumilla
- II Chattogram
- III Noakhali
- IV Barishal
- V Sundarbans belt
- VI Sylhet haors
- VII Sirajganj
- VIII Northern chars

ARKGO

Made in Bangladesh, by design

1

Truss weld shop

Jig-welded frames, local skills

2

Hull molding

Reusable molds, boatyard know-how

3

Powertrain

Imported motor & cells, local assembly

4

Medical fit-out

Stretcher, O2, monitor to DGHS spec

5

QC & float test

IP test, tank float, road + water trial

~70%

local content by value

5 → 20

units / month within 24 months

US$110K

capex for the first shop

Partners: light-engineering clusters (Bogura, Jashore, Dholaikhal) · FRP boatyards (Barishal) · battery & motor importers (Dhaka)

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120 units and ~US$1M revenue in 3 years

Q1 · Build & tank-test

Rev-2 prototype, float & seal tests

Q2 · Trials & paperwork

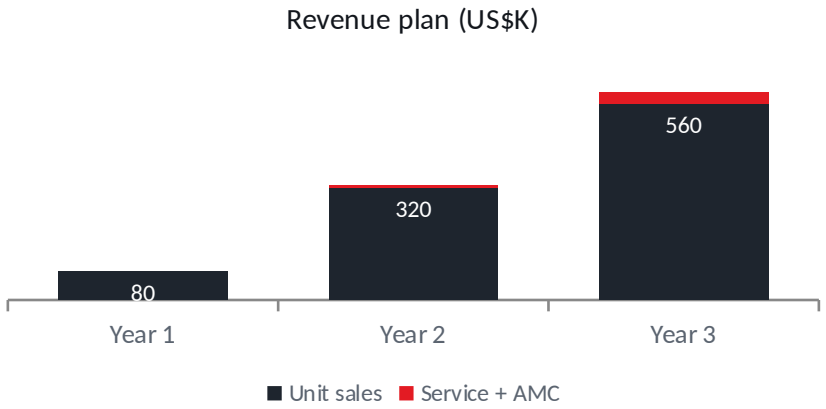
Road + river trials, BRTA filings

Q3 · Monsoon pilot

2 units live in the Sylhet haors

Q4 · First revenue

First e-GP bids, 5-unit batch



YEAR 1 · 10 units	Pilot districts, certification complete
YEAR 2 · 40 units	Fire Service framework + 2 NGO fleets
YEAR 3 · 70 units	Includes the first 10-unit export order

THE ASK

US$275K seed

to build, certify and pilot ARKGO

60%	Prototype → 5-unit pilot fleet	US$165K
15%	Certification & testing	US$41K
15%	Engineering team	US$41K
10%	Go-to-market & pilots	US$28K

Every monsoon, millions of Bangladeshis are cut off by water. We build the vehicle that refuses to stop at the shoreline.

“In water, in land — your life is in our hand.”

